

**BIOGRAPHICAL SKETCH**

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NAME: Garcia Garcia, Jesus

eRA COMMONS USERNAME (credential, e.g., agency login): J.GGARCIA

POSITION TITLE: Senior AI/HPC Systems Engineer

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

| INSTITUTION AND LOCATION                 | DEGREE (if applicable) | START DATE MM/YYYY | COMPLETION DATE MM/YYYY | FIELD OF STUDY                                                                                                                                           |
|------------------------------------------|------------------------|--------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| University of Minnesota, Minneapolis, MN | BS                     | 09/2015            | 05/2018                 | Computational Biology, Physiology, and Psychology                                                                                                        |
| University of Minnesota, Minneapolis, MN | OTH                    | 09/2019            | 06/2021                 | Master's Thesis (Transferred to PhD): Computational Biology, focused on single cell genomics and artificial intelligence applications in cancer research |
| University of Minnesota, Minneapolis, MN | PHD                    | 06/2021            | 06/2027                 | Computational Biology, focused on single cell genomics and artificial intelligence applications in cancer research                                       |

**A. Personal Statement****[Professional Background and Current Role]**

I am a computational biologist and AI / high-performance computing systems engineer focused on supporting large scale cancer genomics and translational research through advanced computing infrastructure and data driven analysis. I currently serve as a Senior AI/HPC Systems Engineer at Mayo Clinic, where I design, deploy, and optimize next generation AI and computing systems that support clinical cancer research. My work enables large scale analysis of genomic, imaging, and multimodal clinical data on NVIDIA Superpod and Blackwell based systems. In this role, I work closely with physician scientists, clinical researchers, and computational teams to ensure that advanced computing technologies are reliable, secure, and scalable for use in patient focused research environments. My responsibilities include supporting research workflows, improving system performance, and enabling the practical use of AI and data intensive methods in clinical research. Previously, I worked at the University of Colorado Boulder, where I supported and expanded the Alpine and Blanca AI supercomputers used for large scale biomedical, scientific computing, and machine/deep learning research across multiple academic disciplines. Before joining Mayo Clinic, I served as a Linux DevOps Engineer at the University of Minnesota Supercomputing Institute, where I supported biomedical research computing and helped deploy and validate a proof-of-concept instance of the LORIS data platform. This platform supported large neurodevelopmental research datasets, including data from the Adolescent Brain Cognitive Development study, one of the largest longitudinal brain studies in the United States. This work was presented to leadership at the National Institutes of Health and demonstrated the importance of centralized computing systems for multi-institution research.

**[Research Focus and Long-Term Goals]**

In parallel with my professional roles, my doctoral research focuses on computational cancer genomics, with an emphasis on brain tumors. I develop computational pipelines to analyze single cell and bulk sequencing data to study tumor cell states, biological pathways, and disease progression. My long-term goal is to continue advancing computational methods and research infrastructure that directly support cancer research, enable collaboration between clinicians and scientists, and improve patient outcomes in the United States.

<https://venteicherlab.umn.edu/members/jesus-garcia-garcia>

## B. Positions and Honors

### Positions and Scientific Appointments

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| 2025 -      | Senior AI/HPC Systems Engineer, Mayo Clinic - Information Technology, Rochester, MN                                   |
| 2024 - 2025 | AI/HPC Systems Engineer, University of Colorado - Research Computing, Boulder, CO                                     |
| 2022 - 2024 | Linux Systems DevOps Engineer, University of Minnesota - Minnesota Supercomputing Institute, Minneapolis, MN          |
| 2021 -      | PhD Candidate, University of Minnesota - Neurosurgery, Minneapolis, MN                                                |
| 2021 - 2022 | Software Packaging Engineer, University of Minnesota - Office of Information Technology, Minneapolis, MN              |
| 2017 - 2021 | Hematology, Oncology and Transplantation Researcher, University of Minnesota - Masonic Cancer Center, Minneapolis, MN |
| 2016 - 2021 | Desktop Support Technician, University of Minnesota - Office of Information Technology, Minneapolis, MN               |
| 2013 - 2015 | Science, Health & Environment Researcher, Minnesota State University - Psychology, Moorhead, MN                       |

### Honors

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| 2018 - 2025 | The Regents Scholarship Program, University of Minnesota                  |
| 2025        | Enhancing Data Science Outcomes with Efficient Workflow, NVIDIA           |
| 2025        | Generative AI with Diffusion Models, NVIDIA                               |
| 2024        | L'Space Mission Concept Academy, NASA                                     |
| 2024        | Transform to Open Science (TOPS), NASA                                    |
| 2024        | International Space Apps Mentor, NASA                                     |
| 2018        | Outstanding Abstract Achievement Award, American Society of Hematology    |
| 2018        | President's Student Leadership and Service Award, University of Minnesota |

## C. Contribution to Science

- [Computational study of chordoma and skull base tumors]** My research has contributed to the study of chordoma and other skull base tumors using computational genomics. Chordoma is a rare cancer with few recurrent coding mutations, which makes standard gene-based analysis difficult. To address this, I developed computational workflows that combine bulk RNA sequencing, single cell RNA sequencing, epigenomic data, and spatial transcriptomics. These analyses focus on tumor cell states, pathway activity, and disease progression rather than single genes. By using high performance computing systems, I was able to analyze large datasets and support a systems level understanding of tumor biology that can guide future therapeutic studies. <https://venteicherlab.umn.edu/research>
- [Studying the developmental origin of chordoma using multi omics data]** I contributed to the study of the developmental origin of chordoma by integrating multi omics data from mouse developmental models and human tumor samples. By analyzing gene expression and regulatory patterns from embryonic notochord tissues in mice and comparing them with bulk and single cell data from human chordoma tumors, I identified shared biological programs linked to early development. This work supports the idea that chordoma tumors retain features of their developmental origin rather than arising from random genetic mutations. Understanding how developmental programs are reused in cancer provides insight into why chordoma behaves differently from other cancers and helps identify biological processes that may be targeted for future therapies.
- [Deployment of AI and computing infrastructure for clinical cancer research at Mayo Clinic]** At Mayo Clinic, I have contributed to science by supporting the deployment and use of advanced AI and computing infrastructure for clinical and translational research. I work on systems based on NVIDIA Superpod and Blackwell architecture that support large scale analysis of genomic, imaging, and multimodal clinical data. My role focuses on enabling researchers and physician scientists to run AI and data analysis workflows reliably in a clinical research environment. This work helps accelerate the development and evaluation of

AI methods for cancer research and supports Mayo Clinic's efforts to integrate advanced computing into patient focused studies. <https://newsnetwork.mayoclinic.org/discussion/mayo-clinic-deploys-nvidia-blackwell-infrastructure-to-drive-generative-ai-solutions-in-medicine/>

4. **[Support of large biomedical research studies through shared data platforms]** At the University of Minnesota Supercomputing Institute, I contributed to the deployment of a proof-of-concept instance of the LORIS data platform in collaboration with the Masonic Institute for the Developing Brain. This effort supported the management and analysis of data from the Adolescent Brain Cognitive Development study, a large multi-site research project. My work focused on enabling secure access, remote analysis, and consistent data organization across institutions. This project was presented to leadership at the National Institutes of Health and demonstrated how shared computing platforms can support large biomedical studies. <https://midb.umn.edu/research/informatics>
5. **[Clinical outcomes research in hematopoietic cell transplantation]** Earlier in my career, I contributed to clinical outcomes research in hematopoietic cell transplantation by co-authoring a study that examined long term healthcare use among adult transplant recipients. Using data from 1,077 patients treated at the University of Minnesota between 2000 and 2016, we compared outcomes across different graft sources. The study showed that patients who received umbilical cord blood transplants had lower healthcare use through five years after transplant compared to those who received bone marrow or peripheral blood stem cell grafts. This work provided evidence to support clinical decision making in transplant medicine and was presented at the American Society of Hematology annual meeting, where it received the Outstanding Abstract Achievement Award. <https://ashpublications.org/bloodadvances/article/5/3/853/475145/Low-5-year-health-care-burdenafter-umbilical-cord>

## D. Scholastic Performance

### Scholastic Performance

| YEAR                                    | COURSE TITLE                             | GRADE |
|-----------------------------------------|------------------------------------------|-------|
| UNIVERSITY OF MINNESOTA (UNDERGRADUATE) |                                          |       |
| 2016                                    | Orientation to Health Careers            | A     |
| 2016                                    | The Future Physician I                   | A     |
| 2016                                    | Foundations of Biology II Lab            | A     |
| 2016                                    | Molecular Biology and Society            | S     |
| 2017                                    | Medical Terminology                      | A     |
| 2017                                    | Calculus I                               | A-    |
| 2017                                    | Comp Programming for Biology             | A     |
| 2017                                    | Physiology Majors Paper                  | A-    |
| 2017                                    | Physiology Laboratory                    | A-    |
| 2018                                    | Humanistic Healthcare and Communications | A     |
| 2018                                    | Principles of Physiology                 | B     |
| UNIVERSITY OF MINNESOTA (GRADUATE)      |                                          |       |
| 2019                                    | Biostatistics I                          | A     |
| 2019                                    | Ethics in Bioinformatics                 | A     |
| 2019                                    | Intro to Computational for Biologists    | B     |
| 2019                                    | Computational Techs. Genomics            | A     |
| 2020                                    | Computation and Biology                  | A     |
| 2020                                    | Fun. Gen., Sys. Bio, & Bioinf.           | A     |
| 2020                                    | Translational Bioinformatics             | A     |
| 2020                                    | Translation Bioinformatics Lab           | A     |
| 2021                                    | Integrative Genomics                     | A     |
| 2022                                    | Doctorate Thesis Credits (Completed)     | NG    |

In the United States, academic evaluation typically employs a system of five, six, or seven-letter grades. The conventional grading scale consists of A+, A, A-, B+, B, B-, C+, C, C-, D+, D, D-, and F, where A+ represents the highest achievement and F indicates the lowest. Occasionally, numerical grades may be used. The conversion of numeric grades to letter grades may vary across institutions, disciplines, and academic levels.